# Topological Quantum Field Theory and the Jones Polynomial

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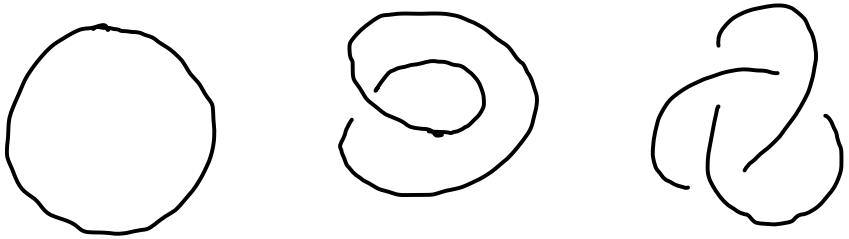
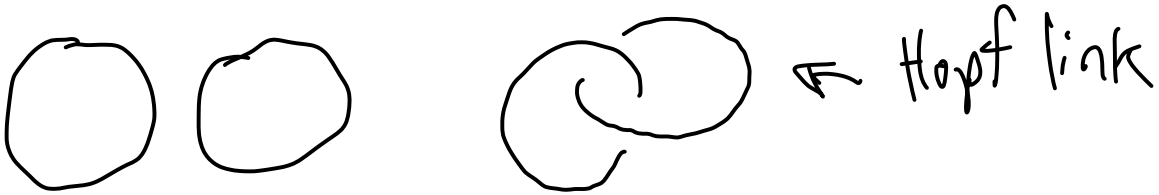
# Knots & Links

A **knot** is an equivalence class of embeddings $K : S^1 \to S^3$ considered up to ambient isotopy.



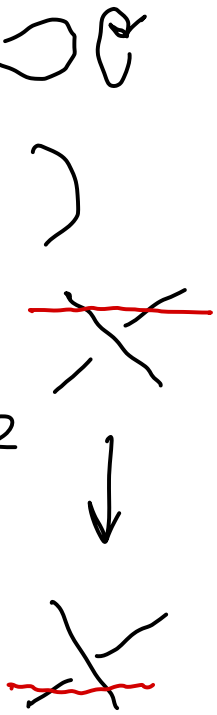
A **link** is a disjoint union of knots considered up to ambient isotopy.



# Topological Invariants

Given I have two knots or links, how does one determine whether they are distinct?

If I have two knot diagrams: $K_1$ and $K_2$. There's a sequence of "Reidemeister moves" takes $K_1 \to K_2$



As in general topology, we distinguish knots/links by means of **topological invariants**.

Homology, $\pi_1(X)$, cohomology

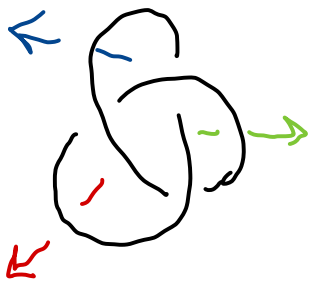
# Knot Invariants?

Ordinary (co)homology is useless for knots!

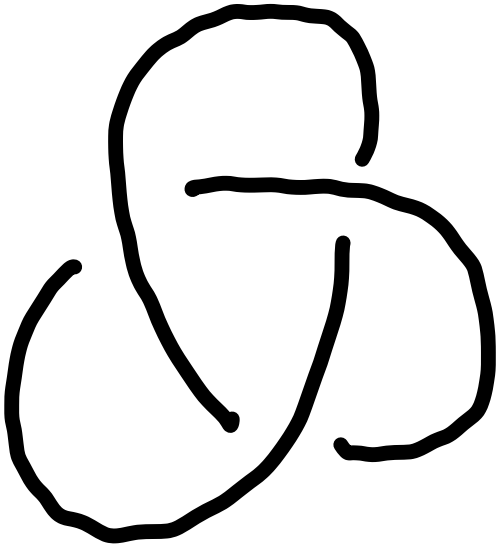
Alexander duality $\tilde{H}_k(S^n - X) \cong \tilde{H}^{n-k-1}(X)$

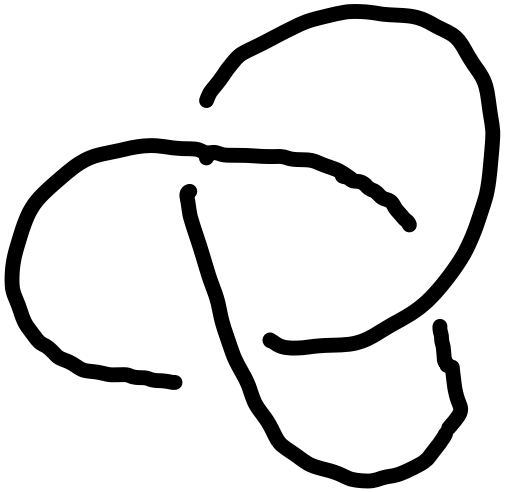
$X = K \cong S^1$

The fundamental group of the complement of your knot/link is useful, but far from perfect.

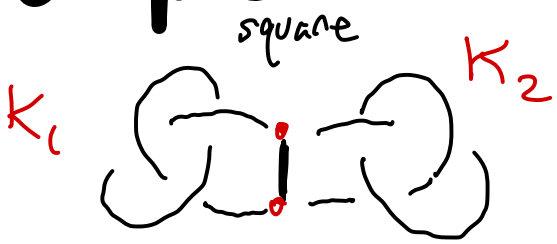


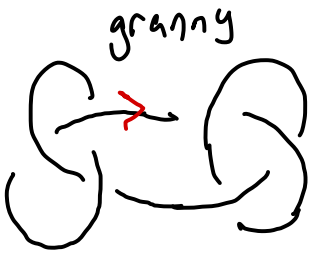
# Fundamental group cannot detect knot parity





A knot is "prime" if it can't be decomposed as connected sum

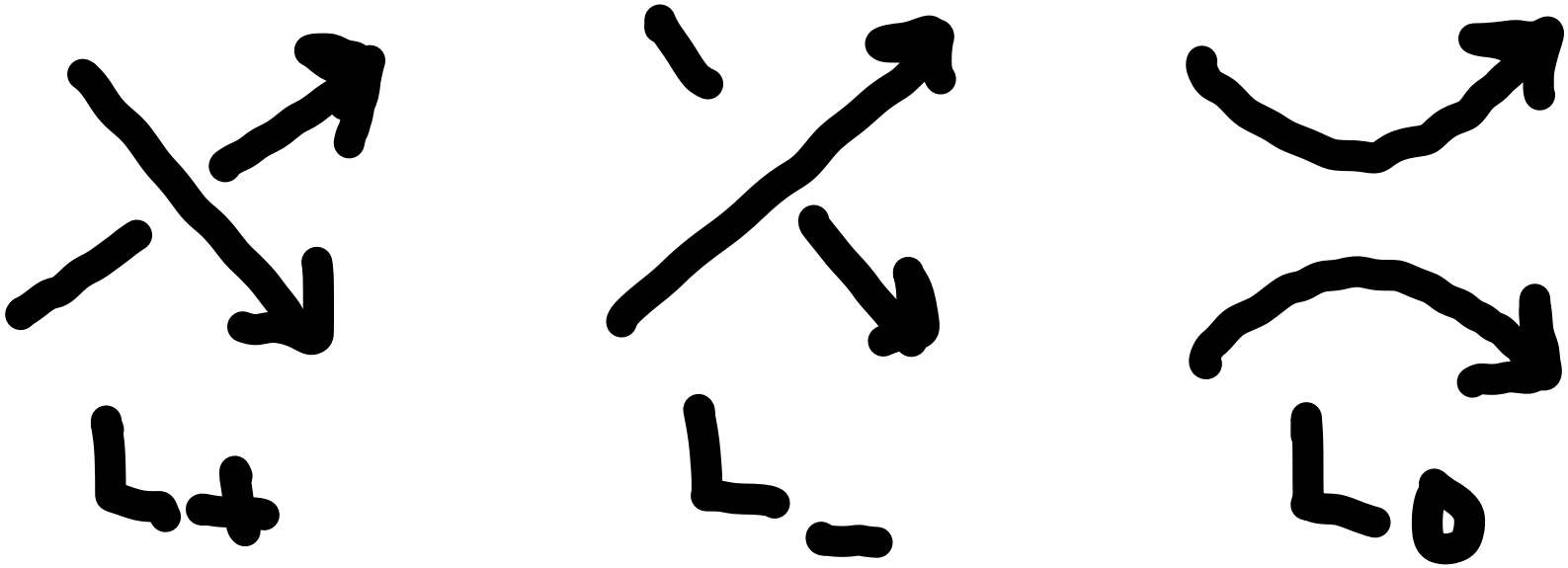


# Some mad science...

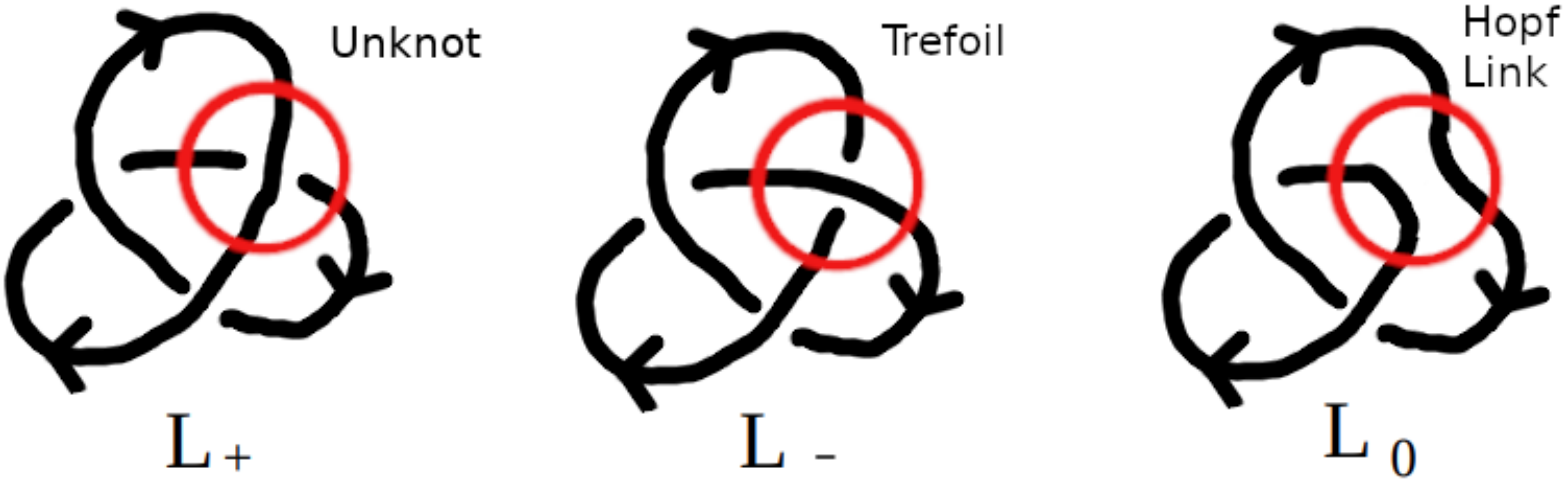
Given a link $L$, suppose we wanted to assign a polynomial $J_L$ to it. How to get started?

Let's give $L$ an orientation, choose a crossing, and play "mad scientist" by replacing it with one of the choices:



# "Franken-links"

Picking the right-most crossing in the (left-handed) trefoil, we can create the following "Franken-links":




We'll use these "Franken-links" to define what's known as a **skein relation** that determines the knot polynomials.

# The Jones polynomial

$$\mathcal{L} = \{\text{all oriented (tame) links in } \mathbb{R}^3\}$$

$$J : \mathcal{L} \longrightarrow \mathbb{Z}[t, t^{-1}]$$

$$J : L \longmapsto J_L$$

- $L_1 \simeq L_2$, $J_{L_1} = J_{L_2}$
- $L =$ unknot, $J_L = 1$
- $t^{-1} \cdot P_{L_+} - t \cdot P_{L_-} = (t^{\frac{1}{2}} + t^{-\frac{1}{2}}) P_{L_0}$

# Some remarks

Why doesn't choice of crossing matter?

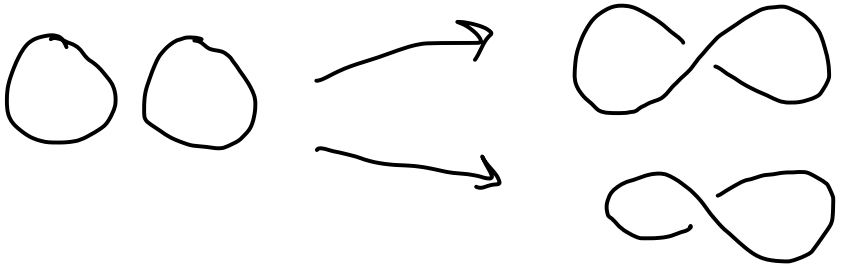
There's an inductive argument using fact true "finitely unknotted".

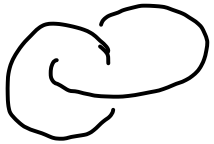
~ Lickorish Chapter 15 HOMFLY

Open problem

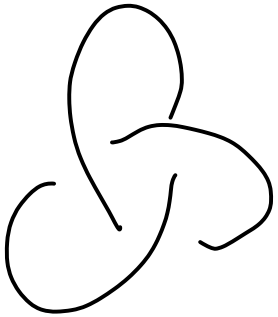
Is there a non-trivial knot with 1 as its Jones polynomial?

# Examples

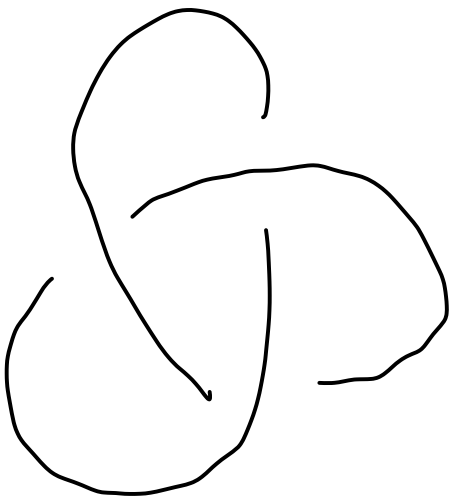
-  $t^{\frac{1}{2}} - t^{-\frac{1}{2}}$

-  $t^{-1} \cdot P_{\text{Hopf Link}} - t\, P_{OO} = (t^{\frac{1}{2}} - t^{-\frac{1}{2}}) P_O$

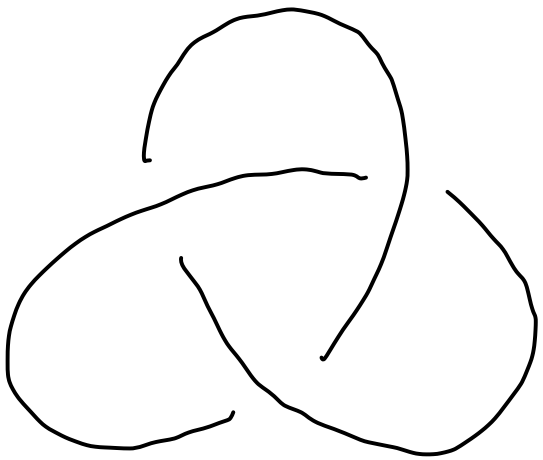
  $\rightsquigarrow t^{\frac{1}{2}}(1 + t^2)$

-  $\rightsquigarrow t^{-1} + t^{-3} - t^{-4}$

# The Jones polynomial detects knot / link parity



Left-handed : $t^{-1}+t^{-3}-t^{-4}$



Right-handed : $t+t^{3}-t^{4}$

# The cobordism category

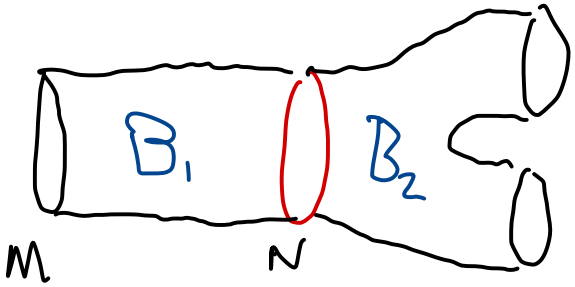
$\mathrm{Cob}(d)$

- objects: are closed, oriented $d$-manifolds
- morphisms: $B : \Sigma_1 \longrightarrow \Sigma_2$

  $B$ is a equivalence class of $(d+1)$-dManifolds

  $\partial B = \overline{\Sigma_1} \sqcup \Sigma_2$ ( $\overline{\Sigma_1} = \Sigma_1$ w/ reversed orientation )
- $B_1, B_2$ are bordisms, $\partial B_1 = \overline{M} \sqcup N$, $\partial B_2 = \overline{N} \sqcup O$

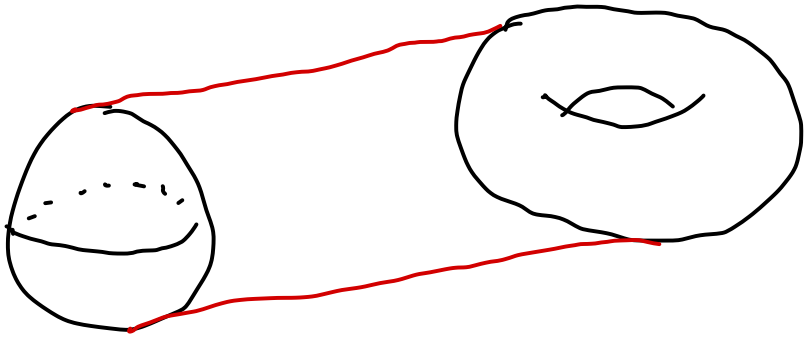
  $B = B_2 \circ B_1 = B_1 \sqcup_{N \sim \overline{N}} B_2$
- Regard disjoint union as a product



# Topological Quantum Field Theory (Atiyah-Segel)

$$Z : \mathrm{Cob}(d) \longrightarrow V_F$$

- $Z(\emptyset) = F$
- $Z(M \sqcup N) = Z(M) \otimes Z(N)$
- $Z(\overline{M}) = Z(M)^*$



# A few interesting remarks

$Z$ not completely determined, but <u>strong</u> constraints on behavior.

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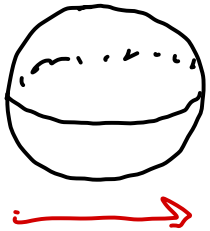
$\Sigma \longrightarrow \Sigma$, $\Sigma \times I$

$Z(\Sigma \times I) \rightsquigarrow i_\Sigma : Z(\Sigma) \longrightarrow Z(\Sigma)$

$$i_\Sigma \circ i_\Sigma = i_\Sigma$$

$B$ is $(d+1)$-manifold and closed.

$\partial B = \emptyset = \emptyset \sqcup \emptyset$ $\emptyset$  $\emptyset$

Closed $(d+1)$-manifolds

$E \longrightarrow F$

# Witten's TQFT

2-dimensional TQFT

- 3-manifolds $\rightsquigarrow$ maps / numbers (if closed)
- 2-manifolds $\rightsquigarrow$ vector spaces
- 1-manifolds $\rightsquigarrow$ vectors

$(B, L, \mu_i)$

closed $(d+1)$-manifold

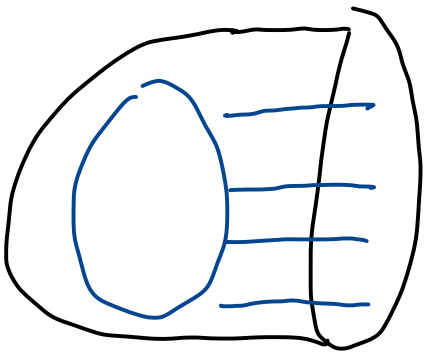
Link $L \subset B$

collection of representations of a compact, simple Lie group one for each component in $L$.

# Witten's TQFT

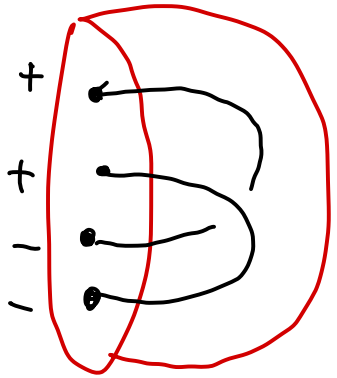
$B = S^3$

$S^3 - B^3$



$L \sim L'$

$B^3$




$L' = L \cap B^3$

$L'$ has reps $\mu_{i_1}, \mu_{i_2}$

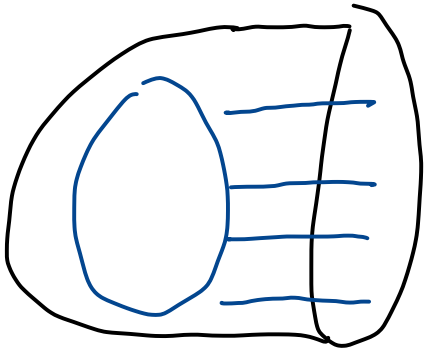
$Z(B, L', \mu_{i_1}, \mu_{i_2}) \rightsquigarrow \chi \in \mathcal{H}$ ← vector space $Z$ associates to $\partial B^3 = S^2$

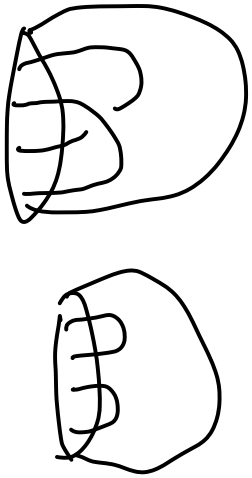
# Witten's TQFT

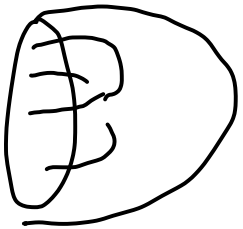
$$Z(S^3 - B, L - L', \mu_i) = \psi \in H^*$$

$$Z(S^3, L, \mu_i) = \psi(X)$$

- Witten showed $\dim(H) = 2$







$$\alpha Z(S^3, L_-, \mu_i) + \beta Z(S^3, L_+, \mu_i) + \gamma Z(S^3, L_0, \mu_i) = 0$$

# Witten's TQFT

# Principle $G$-bundles and Connections

Given a manifold $B$ and a topological group $G$, a $G$-**bundle** is a fiber bundle equipped with a regular action by $G$ (i.e. to each point in the fiber, we can uniquely assign a group element).

Given a $G$ a Lie group, we can generalize the notion of a connection on a vector bundle to create what's known as a **principle connection** on a principle $G$-bundle $B \xrightarrow{\pi} M$, which is a 1-form on $B$ taking values in the Lie algebra $\mathfrak{g}$.

In a nut-shell, right multiplication by $g$ induces a linear transformation on the Lie algebra and this introduces a relationship between objects in one fiber and objects in nearby fibers.

# Holonomy of a connection

Letting $\pi : P \to M$ be a principle $G$-bundle and $\omega$ a principle connection, then given a smooth loop $\gamma : [0,1] \to M$ starting and ending at $m \in M$, the connection yields a unique lifting to the principle $G$-bundle $\widetilde{\gamma} : [0,1] \to P$ which starts a particular $p \in \pi^{-1}(m)$ and ends at some $p \cdot g \in \pi^{-1}(m)$.

We can then define

$$Hol_p(\omega) = \{g \in G \mid p \sim p \cdot g\}$$

# Wilson loop

Given some $K \subset M$ be some closed curve,

$$W_r(C) = Tr_R P \; exp \int_C A_i dx^i$$

The invariants are given by the "vacuum expectation" vvalues of products of these Wilson loops:

$$Z(M, L, \mu_i) = \int D\mathcal{A} exp(i\mathcal{L}) \prod_{i=1}^{r} W_{R_i}(C_i)$$

# Thank you!

Special thanks to the organizers and to my collaborator:

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